

[illegible]

$$\mathcal{H}_A = -\frac{U}{2} (\hat{n}_{d\uparrow} - \hat{n}_{d\downarrow})^2 + \sum_{\vec{k}, \sigma} \epsilon_{\vec{k}} \tau_{\vec{k}, \sigma} + V \sum_{\sigma} \left(c_{d\sigma}^\dagger c_{0\sigma} + h.c. \right),$$

$$\begin{array}{l} \tau_{\vec{k},\sigma}^+ \equiv \\ c_{\vec{k},\sigma}^+ c_{\vec{k},\sigma}^- \\ \frac{1}{2} \\ \sum_k c_{k,\sigma}^\dagger c_{k,\sigma} = \\ U \rightarrow \\ T = \end{array}$$